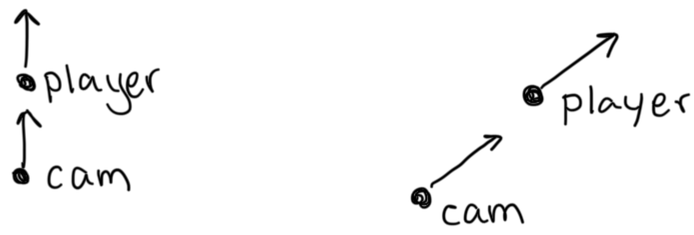
We want the camera to rotate/move as the player rotates/moves.

E.g.



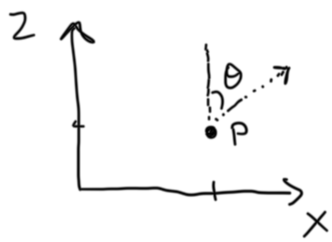
before player rotation

after player rotation

Let: (float) currentRotationAngle = transform.eulerAngles.y

Suppose the player is currently rotated 45°

and is at position (3, 2)
x z



$\theta = 45°$

Set the position of the camera:

```
transform.position =
  new Vector3(target.position.x, target.position.y + 3.0f, target.position.z)
```

Where target is the player's transform.
This way the cam is at the same x, z coordinates as the player but 3 units above the player.

Transform the current rotation from float to Quaternion

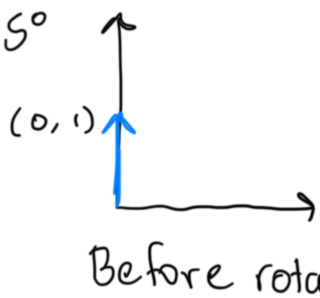
```
Quaternion currentRotation = Quaternion.Euler(0, currentRotationAngle, 0)
```

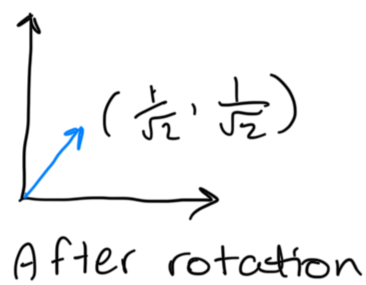
Then we can use this quaternion to start getting the position of the camera.

```
Vector3 rotatedPosition = currentRotation * Vector3.forward
```

↳ This rotates Vector3.forward this many degrees around the y-axis

E.g. $\theta = 45°$



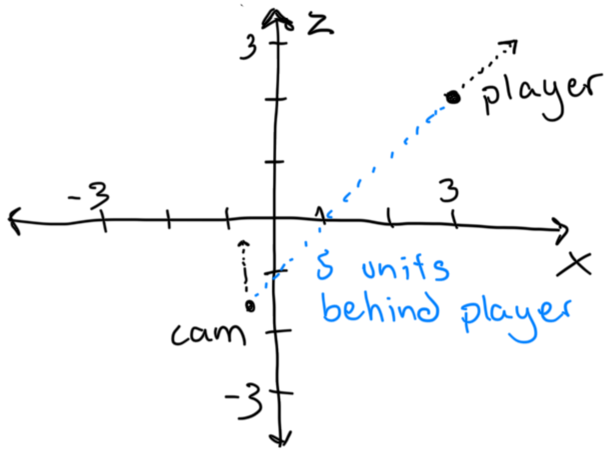


Then we can get the final position of the camera. This blue vector is facing the same direction as the player. So to get the camera behind the player we will subtract (a multiple of) it from the player position.

```
transform.position -= rotatedPosition * 5
```

so cam is 5 units behind the player

E.g.



$$\text{transform.position} = (3,2) - \left(\tfrac{1}{\sqrt{2}}, \tfrac{1}{\sqrt{2}}\right)\cdot 5 \approx (-0.5, -1.5)$$

Point camera in the direction of the player

At this point, the cam is not facing the player yet.

We can do this simply by using Unity's built in method:

$$\underbrace{\text{transform}}_{\text{cam transform}}\text{.LookAt(}\underbrace{\text{target}}_{\text{transform to rotate towards/look at}}\text{)}$$